Task : Bash Scripting

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1. Objective -

The objective of this task was to write a Bash script inside `read_secret.sh` that reads the contents of `secret.txt`, stores that content in a variable, and prints the value of that variable to the terminal which reveals the contents of `secret.txt`.

2. Prerequisites – Why Use a Variable Instead of Just `cat`?

This assignment specifically requires storing the file's content in a variable before printing it, rather than simply running `cat secret.txt` directly. This is **command substitution** - a core Bash scripting concept where the output of a command is captured and assigned to a variable using the `$(...)` syntax.

3. Initial Steps

Before writing the script, I explored the working directory to understand the file structure and permissions I was working with :

```
student@hacking:~$ ls
loot  practice  read_secret.sh  run_secret
secret.txt
```

Running `ls -la` on the home directory showed the following permissions:

```
-rwxrwxrwx 1 root   student   102  read_secret.sh
-rwsr-xr-x 1 root   root    16008  run_secret
-rw-r----- 1 root   perm       26  secret.txt
```

Interpretation of these permissions:

- `secret.txt` — owned by root, group perm. Permissions are rw-r-----, meaning only the owner (root) can read/write it, and only members of the perm group have read access. As the student user, I belong to neither, so direct access is denied.

- `read_secret.sh` — owned by root but it is readable,writable and executable (rwxrwxrwx), meaning any user, including student, can edit its contents.
- `run_secret` — owned by root:root with permissions `rwsr-xr-x.` The s in place of the owner's execute bit indicates the **SUID (Set User ID)** bit is set. This means that whenever this binary is executed, it runs with the privileges of its owner (root) regardless of which user actually runs it.

4. Step-by-Step Process

Step 1: Attempting Direct Access

As expected from the permission analysis above, direct access as the student user was denied, I also attempted to work around this by changing the file's permissions directly:

```
student@hacking:~$ cat secret.txt
cat: secret.txt: Permission denied
student@hacking:~$ chmod +r secret.txt
chmod: changing permissions of 'secret.txt':
Operation not permitted
```

Step 2: Editing `read_secret.sh`

Since `read_secret.sh` was writable, I opened it in a text editor and wrote the following script :

```
student@hacking:~$ nano read_secret.sh
#!/bin/bash
content=$(cat secret.txt)
echo "$content"
```

Interpretation :

- `#!/bin/bash` - the shebang line, which tells the system to interpret this script using the Bash shell.
- `content=$(cat secret.txt)` - this runs cat secret.txt and captures its output, storing it inside the variable content using command substitution `($(...))`.
- `echo "$content"` prints the value stored in the variable to the terminal.

I saved the file and exited the editor (ctrl+X , Y , Enter in nano).

Step 3: Running the Script Directly :

Running `read_secret.sh` directly as the `student` user failed to read `secret.txt,` because a script executes with the privileges of the user who runs it — not the user who owns the script file. Since `secret.txt` was unreadable by `student` regardless of who wrote the script, the `cat` command inside the script still hit a `Permission denied` error.

That is where `run_secret`came in.

Step 4: Executing via run_secret

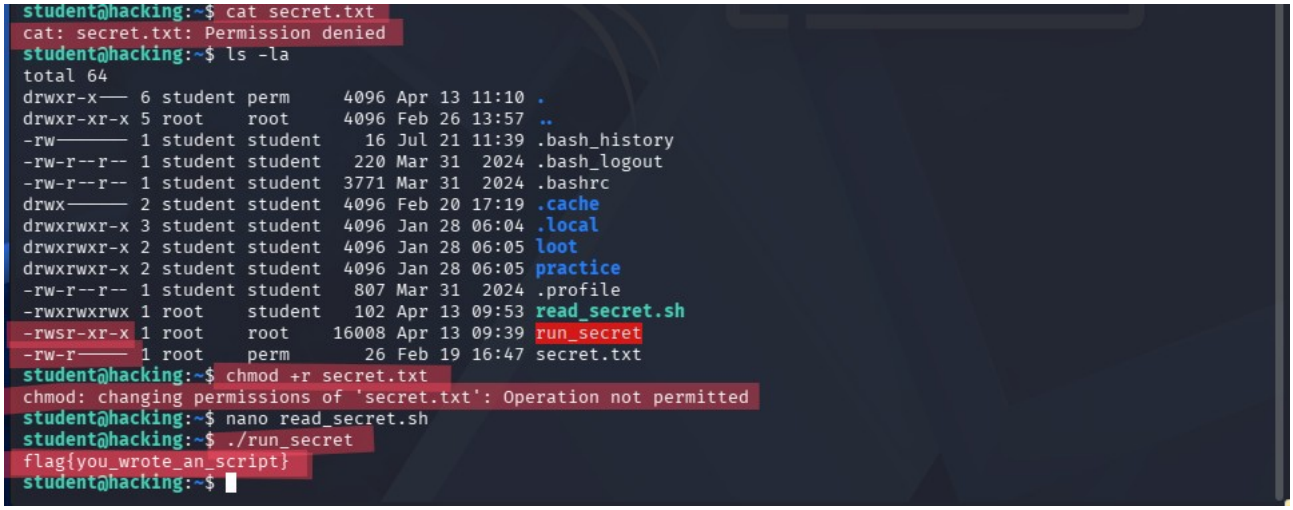
```
student@hacking:~$ ./run_secret
flag{you_wrote_an_script}
```

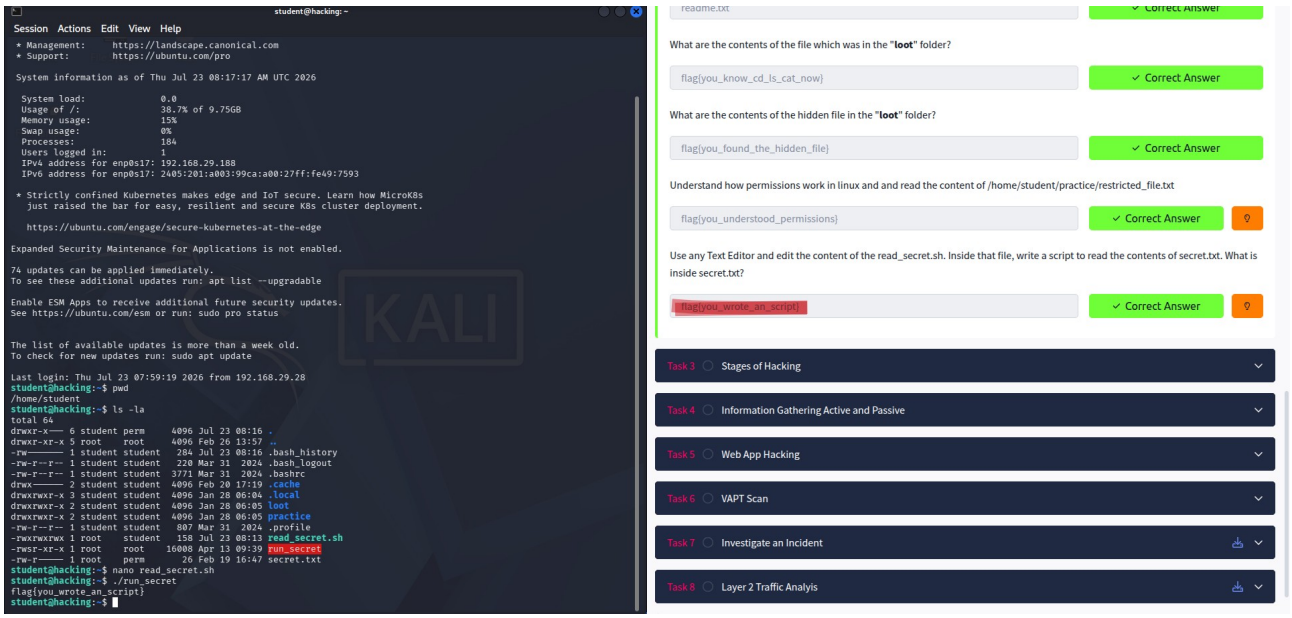
Because `run_secret` has the **SUID bit** set and is owned by `root`, executing it runs its underlying logic with `root` privileges — regardless of the fact that student invoked it. This allowed the process to successfully read `secret.txt` and output its contents.

Contents of `secret.txt`:

```
flag{you_wrote_an_script}
```

Step 5: Screenshot Evidence





The SUID bit and its risks :

The SUID (Set User ID) permission bit allows a program to run with the privileges of its *owner*, rather than the user executing it. This is normally used for legitimate purposes.However, if a SUID-root binary calls out to a script or file that is **writable by a non-root user** as `read_secret.sh` was in this task, it creates a privilege escalation vulnerability. An attacker can modify the writable script, and when the SUID binary executes it, the malicious or modified code runs with root privileges.

Command substitution as a scripting fundamental:

This task reinforced the practical use of `$(...)` for capturing command output into a variable , a pattern used throughout real-world Bash scripting for logging, validation, and conditional logic, rather than just displaying output once.

5 . Conclusion

This task required writing a Bash script to read the contents of a permission-restricted file and print it via a variable. While the script itself was straightforward, the underlying challenge illustrated an important real-world security concept: how a SUID-root binary combined with a world-writable script can be leveraged to bypass file permission restrictions — in this case, revealing the flag `flag{you_wrote_an_script}`. This reinforced both basic Bash

scripting syntax (command substitution, variables, echo) and a foundational Linux privilege escalation concept.